

Software Engineering Mastery Companion Resource Guide

Curated YouTube Channels, Courses, Documentation, Practice Sites and Learning Resources

Stage 1 – Python

CS50P: <https://cs50.harvard.edu/python/>

Corey Schafer: <https://www.youtube.com/@coreyms>

freeCodeCamp Python: <https://www.youtube.com/@freecodecamp>

Stage 2 – Problem Solving

CS50x: <https://cs50.harvard.edu/x/>

NeetCode: <https://www.youtube.com/@NeetCode>

HackerRank: <https://www.hackerrank.com>

Stage 3 – Data Structures

MIT 6.006: <https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/>

Abdul Bari: https://www.youtube.com/@abdul_bari

VisuAlgo: <https://visualgo.net>

Stage 4 – Algorithms

MIT Algorithms:

<https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/resources/lecture-videos/>

CS50 Algorithms: <https://cs50.harvard.edu/x/>

NeetCode: <https://www.youtube.com/@NeetCode>

Stage 5 – C

CS50 C Weeks: <https://cs50.harvard.edu/x/>

Beej's Guide to C: <https://beej.us/guide/>

freeCodeCamp C: <https://www.youtube.com/@freecodecamp>

Stage 6 – C++

learncpp: <https://www.learncpp.com/>

The Chernobyl: <https://www.youtube.com/@TheCherno>

CppReference: <https://en.cppreference.com>

Stage 7 – Git & Linux

Git Official Docs: <https://git-scm.com/doc>

Missing Semester: <https://missing.csail.mit.edu/>

NetworkChuck Linux: <https://www.youtube.com/@NetworkChuck>

Stage 8 – Operating Systems

OSTEP: <https://pages.cs.wisc.edu/~remzi/OSTEP/>

Computerphile: <https://www.youtube.com/@Computerphile>

MIT Systems: <https://ocw.mit.edu>

Stage 9 – Networking

Beej's Networking Guide: <https://beej.us/guide/bgnet/>

Practical Networking: <https://www.youtube.com/@PracticalNetworking>

Computer Networking Course: <https://www.youtube.com/@freecodecamp>

Stage 10 – Databases

PostgreSQL Docs: <https://www.postgresql.org/docs/>

Database Design: <https://www.youtube.com/@freecodecamp>

SQLBolt: <https://sqlbolt.com>

Stage 11 – Backend APIs

FastAPI Docs: <https://fastapi.tiangolo.com/>

Flask Docs: <https://flask.palletsprojects.com/>

freeCodeCamp APIs: <https://www.youtube.com/@freecodecamp>

Stage 12 – Testing & Debugging

pytest Docs: <https://docs.pytest.org/>

Debugging Guide: <https://realpython.com/>

Corey Schafer Testing: <https://www.youtube.com/@coreyms>

Stage 13 – System Design

System Design Primer: <https://github.com/donnemartin/system-design-primer>

Gaurav Sen: <https://www.youtube.com/@gkcs>

ByteByteGo: <https://www.youtube.com/@ByteByteGo>

Stage 14 – Docker & Cloud

Docker Docs: <https://docs.docker.com/>

Docker Course: <https://www.youtube.com/@freecodecamp>

AWS Skill Builder: <https://skillbuilder.aws>

Stage 15 – Portfolio Projects

GitHub: <https://github.com>

Awesome Projects: <https://github.com/topics>

freeCodeCamp Projects: <https://www.freecodecamp.org>

Stage 16 – Job Readiness

Tech Interview Handbook: <https://www.techinterviewhandbook.org/>

NeetCode: <https://www.youtube.com/@NeetCode>

Pramp: <https://www.pramp.com>

Stage 17 – Specialization

roadmap.sh: <https://roadmap.sh>

OSSU: <https://github.com/ossu/computer-science>

freeCodeCamp: <https://www.freecodecamp.org>

Recommended Core Channels

- CS50
- freeCodeCamp
- Corey Schafer
- The Cherno
- Computerphile
- Abdul Bari
- Gaurav Sen
- ByteByteGo
- Programming with Mosh
- Tech With Tim